

Ideation Phase

Brainstorm & Idea Prioritization Template

Date	1 July 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	4 Marks

Brainstorm & Idea Prioritization

Project: Rising Waters – Intelligent Flood Prediction and Early Warning System

Floods can cause serious damage to people, property, and the environment. Our team discussed different ideas to find the best solution for this problem. During the brainstorming session, we explored ideas such as flood prediction, weather analysis, early warning systems, and disaster management. We compared these ideas based on their impact and feasibility. After evaluating all the ideas, we selected **Rising Waters**, a Machine Learning-based flood prediction system. It uses historical weather and rainfall data to predict flood risk and provide early warnings. The system helps authorities plan evacuations, manage resources, and improve disaster preparedness through an easy-to-use Flask web application.

This brainstorming activity helped our team choose the most practical and effective solution for our project.

Step-1: Team Gathering, Collaboration and Select the Problem Statement



Brainstorm & Idea Prioritization

Machine Learning-Based Flood Prediction System





RAINFALL MONITORING



WATER LEVEL ANALYSIS



MACHINE LEARNING PREDICTION



EARLY WARNING ALERTS

Predict. Prepare. Protect.
Smarter Predictions for a Safer Tomorrow.

1

Before You Collaborate

A little preparation leads to better ideas and stronger solutions.

- 1 Team Gathering**
 - ✓ Invite a diverse group of team members.
 - ✓ Clarify roles and responsibilities.
 - ✓ Ensure everyone has a voice.
 - ✓ Create a safe and open environment.
- 2 Set the Goal**
 - ✓ Define what success looks like.
 - ✓ Align on the purpose of this session.
 - ✓ Identify what we want to achieve.
 - ✓ Keep the goal specific and measurable.
- 3 Learn About the Problem**
 - ✓ Understand the background and context.
 - ✓ Review existing data and research.
 - ✓ Identify pain points and root causes.
 - ✓ Gather insights from key stakeholders.

2

Define Your Problem Statement

A well-defined problem leads to meaningful ideas.

PROBLEM STATEMENT

How might we develop a **Machine Learning-based Flood Prediction System** that accurately predicts flood risk using historical weather and rainfall data to provide early warnings and support disaster management authorities?



KEY RULES OF BRAINSTORMING

- 1. STAY ON TOPIC**
Focus on the problem and the goal.
- 2. ENCOURAGE IDEAS**
The more ideas, the better.
- 3. DEFER JUDGMENT**
Hold evaluations until later.
- 4. LISTEN TO OTHERS**
Be open and build on ideas.
- 5. THINK VISUALLY**
Sketch, diagram, and use visuals.
- 6. COLLABORATE**
Work together to create better solutions.

3

Rising Waters: Intelligent Flood Prediction and Early Warning System

Step 2: Brainstorm – Idea Listing

Each person contributes ideas independently.

Diagram illustrating the brainstorming phase where six individuals contribute ideas independently. The ideas are organized into six categories, one for each person:

- Person 1:** Historical Weather Data, Rainfall Analysis, Cloud Cover, Humidity.
- Person 2:** Machine Learning Model, Decision Tree, XGBoost, Feature Engineering.
- Person 3:** Flood Prediction, Early Warning, Real-Time Alerts, Threshold Levels.
- Person 4:** Flask Web App, Dashboard, User Interface, Visualizations.
- Person 5:** Disaster Management, Resource Allocation, Evacuation Planning, Risk Assessment.
- Person 6:** Temperature, Wind Speed, Humidity, Atmospheric Pressure.
- Person 7:** Weather Analytics, Data Preprocessing, Time Series Analysis, Model Evaluation.
- Person 8:** Real-Time Alerts, SMS/Email Notifications, API Integration, Cloud Deployment.

“

Great ideas start when everyone contributes.

”

Step 3: Group Ideas

We cluster similar ideas to build a stronger solution.

Diagram illustrating the clustering phase where similar ideas are grouped to build a stronger solution. The ideas are organized into five main categories:

- Machine Learning:** Machine Learning Model, Decision Tree, XGBoost, Model Evaluation.
- Weather Data:** Historical Weather Data, Rainfall Analysis, Cloud Cover, Humidity, Temperature, Wind Speed.
- Disaster Management:** Disaster Management, Resource Allocation, Evacuation Planning, Risk Assessment.
- User Interface:** Flask Web App, Dashboard, Visualizations, Real-Time Alerts.
- Future Enhancements:** Real-Time Alerts, SMS/Email Notifications, API Integration, Cloud Deployment.

The central focus is the **Intelligent Flood Prediction** system, which integrates these components.

Better data. Smarter predictions. Safer communities.

3 Step 3: Idea Prioritization

Project: *Rising Waters: Intelligent Flood Prediction and Early Warning System*

Our team should all be on the same page about what's important moving forward. Place your ideas on this grid to determine which ideas are important and which are feasible.

 20 minutes

